

# GAF: Gaussian Avatar Reconstruction from Monocular Videos via Multi-view Diffusion

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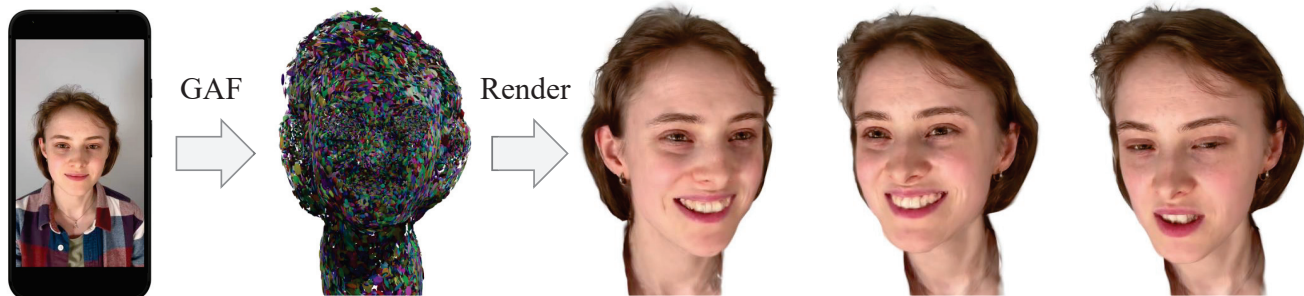


Figure 1. Given a short, monocular video captured by a commodity device such as a smartphone, GAF reconstructs a 3D Gaussian head avatar, which can be re-animated and rendered into photo-realistic novel views. Our key idea is to distill the reconstruction constraints from a multi-view head diffusion model in order to extrapolate to unobserved views and expressions.

## Abstract

We propose a novel approach for reconstructing animatable 3D Gaussian avatars from monocular videos captured by commodity devices like smartphones. Photorealistic 3D head avatar reconstruction from such recordings is challenging due to limited observations, which leaves unobserved regions under-constrained and can lead to artifacts in novel views. To address this problem, we introduce a multi-view head diffusion model, leveraging its priors to fill in missing regions and ensure view consistency in Gaussian splatting renderings. To enable precise viewpoint control, we use normal maps rendered from FLAME-based head reconstruction, which provides pixel-aligned inductive biases. We also condition the diffusion model on VAE features extracted from the input image to preserve facial identity and appearance details. For Gaussian avatar reconstruction, we distill multi-view diffusion priors by using iteratively denoised images as pseudo-ground truths, effectively mitigating over-saturation issues. To further improve photorealism, we apply latent upsampling priors to refine the denoised latent before decoding it into an image. We evaluate our method on the NeRSemble dataset, showing that GAF outperforms previous state-of-the-art methods in novel view synthesis. Furthermore, we demonstrate higher-fidelity avatar reconstructions from monocular videos captured on commodity devices. [Project Page: https://tangjiapeng.github.io/projects/GAF](https://tangjiapeng.github.io/projects/GAF)

## 1. Introduction

Creating photorealistic and animatable head avatars has long been a challenge in computer vision and graphics. This is crucial for a vast variety of applications such as immersive telepresence in virtual and augmented reality, computer games, movie effects, virtual classrooms, and videoconferencing. Here, the goal is to generate avatars that can be realistically rendered from any viewpoint, accurately capturing facial geometry and appearance, while also enabling easy animation for realistic head portrait videos depicting various expressions and poses. The democratization of high-fidelity head avatars from commodity devices is a challenge of widespread interest. However, photo-realistic head avatar reconstruction from monocular videos is challenging and ill-posed, due to limited head observations causing a lack of constraints for novel-view rendering.

To create photorealistic, animatable human avatars, researchers have integrated NeRF [49, 50] and Gaussian Splatting (GS) [31] techniques with parametric head models [23, 39, 76], enhancing control over unseen poses and expressions. Approaches like [24, 57, 90, 91] achieve high-quality head reconstructions and realistic animations using dense multi-view datasets, typically captured in controlled studio settings. However, these methods encounter significant limitations with monocular recordings from commodity cameras, such as smartphone portrait videos. Monocular methods [4, 12, 21, 66, 89, 101, 102, 105] aim to reconstruct head avatars from single-camera videos but often rely on

substantial head rotations to capture various angles. They only reconstruct visible regions from input frames, leaving occluded areas incomplete and under-constrained. For instance, as illustrated in Fig. 4, front-facing videos provide limited capture on side regions, leading to visible artifacts when rendering extreme side views.

To this end, we introduce a multi-view head diffusion model that learns the joint distribution of multi-view head images. Given a single-view input image, our model generates a set of view-consistent output images. By leveraging view-consistent diffusion priors on human heads, our approach enables the robust reconstruction of Gaussian avatars, preserving both appearance and identity across novel perspectives. Unlike existing methods [41, 68, 85, 87] that use camera pose as a conditioning factor for viewpoint control, we use the normal map rendered from a reconstructed parametric face model as guidance. The normal maps provide stronger inductive biases, offering more precise and reliable novel view generation for heads. Specifically, we condition our multi-view diffusion process on 2D features extracted from the input image’s autoencoder, rather than only using high-level semantic vectors like CLIP embeddings [58]. This design allows us to incorporate fine-grained identity and appearance details directly into the multi-view denoising process, ensuring that the generated images maintain coherence and consistency across viewpoints in terms of identity and appearance.

To exploit multi-view diffusion priors for Gaussian head reconstruction, we employ a diffusion loss that uses iteratively denoised images as pseudo-ground truths for novel view renderings, instead of using a single-step score distillation sampling loss [55]. Compared to NeRF, 3D-GS enables real-time multi-view rendering, significantly reducing optimization time. Moreover, to improve the fidelity of the Gaussian renderings, we introduce a latent upsampler model to enhance facial details within the denoised latent before decoding it back to image space. As demonstrated in Fig. 1, our approach generates high-quality, photorealistic novel views of head avatars using only short monocular videos captured by a smartphone. Extensive comparisons with state-of-the-art methods show that our approach delivers higher fidelity and more view-consistent avatar reconstructions from monocular videos. Our GAF significantly outperforms state-of-the-art methods in novel view and expression synthesis on the NeRSemble dataset and other monocular datasets captured by commodity devices. Our contributions can be summarized as follows:

- We introduce a novel approach for reconstructing photorealistic, animatable head avatars from monocular videos with limited coverage captured on commodity devices, using multi-view head diffusion priors to regularize and complete unobserved regions.
- We propose a multi-view head diffusion model that gener-

ates consistent multi-view images from a single-view input, guided by normal maps rendered from FLAME head reconstructions to improve viewpoint control.

- We present a strategy to enhance the photorealism and cross-view consistency of Gaussian splatting by integrating latent upsampling and multi-view diffusion priors.

## 2. Related Work

**3D Scene Representations.** Neural radiance fields [49] and its variants [5–7, 11, 13, 20, 50] revolutionized 3D scene reconstruction from multi-view images. However, NeRF-based methods are often hindered by computational inefficiency during both training and inference stages. Gaussian Splatting [31] represents a scene as a composition of discrete geometric primitives, *i.e.* 3D Gaussians, and employs an explicit rasterizer for rendering. Compared to NeRF, GS has achieved notable runtime speedups in the training and inference stages. This enables real-time performance and more favorable image synthesis. Unlike polygonal meshes, which require careful topology handling, GS supports substantial topology changes, making it more adaptable to complex surfaces and varying densities.

**Parametric Face/Head Models.** Based on statistical priors of 3D morphable models (3DMM)[8, 27, 39, 81, 84], many works [19, 26, 32, 80] learn 3D face/head reconstruction and tracking from single RGB images/videos. More recent methods [23, 76, 99, 101] leveraged signed distance fields [52] and deformation fields [51, 73, 74] based on coordinate-MLPs for more fine-grained geometry reconstruction including hair and beards. HeadGAP [100] and GPHM [92] learned parametric models for head Gaussians. Our work uses the VHAP tracker [1] to obtain coarse geometries as guidance for dynamic avatar reconstruction.

**Photo-realistic Avatar Reconstruction.** To create photo-realistic animatable head avatars, several works [21, 57, 66, 79, 89, 90, 102, 105, 105] have combined NeRF/GS with 3D morphable models (3DMM). Our work is closely related to animatable Gaussian splats [57, 66], which attached splats to the triangles of FLAME mesh, and updated their properties by triangle deformations controlled by FLAME parameters. Although promising results have been achieved, they typically require multi-view setups with high-quality cameras in studio environments [36, 43]. Some works reconstruct avatars from monocular videos [4, 89, 102, 105]. However, they only reconstruct the visible region from inputs, lacking effective priors to complete missing areas. P4D [17], P4D-v2 [18], GAGAvatar [14], and MGGTalk [25] use feed-forward networks to predict expression-driven 3D heads represented by Gaussians or triplane features [53]. While achieving impressive head reenactments, they could exhibit obvious artifacts in novel views. To address this, we utilize multi-view diffusion pri-

ors to complete unobserved regions of the face, ensuring photorealistic renderings from extremely novel viewpoints.

**Distillation 2D Priors for 3D Generation.** Diffusion models [29, 70–72] have been widely applied to 3D asset generation [9, 46, 77, 82, 96, 103]. Some methods leverage large-scale pretrained text-to-image priors [60, 61, 64] for 3D synthesis [40, 75] using score distillation sampling (SDS) loss [55] and its variants [47, 86, 94]. Several studies [10, 28, 104] applied SDS loss for text-driven 3D head avatar generation. For single-view 3D reconstruction, approaches like ReIFusion [47], Magic123 [56], and Dream-booth3D [59] adapt text-to-image diffusion priors to specific objects to preserve identity. However, achieving consistent novel views from text prompts alone remains challenging without leveraging the input image. To address this, we learn multi-view image diffusion priors conditioned on a single image. Rather than using a single-step SDS loss, we use iteratively denoised images as pseudo-ground truths for novel view supervision, similar to ReconFusion [88]. We also introduce pretrained latent upsampler diffusion priors to improve photo-realism of pseudo-ground truths.

**Multi-view Diffusion Models.** Instead of text-to-image diffusion priors, some works learn image-conditioned novel view diffusion priors, which leverage the identity and appearance details of the input image to generate consistent novel views. 3DiM [87] pioneered a diffusion model for pose-conditioned novel view generation. Zero-123 finetuned StableDiffusion [61] on the Objaverse [15, 16] dataset to improve generalization ability. Some works jointly diffused multiple views from a single image via dense 3D self-attention or epipolar attention layers in the denoiser network, including MVDiffusion [78], SyncDreamer [42], Zero-123++ [67], Wonder3D [44], MVDream [68], ImageDream [85], Human3Diffusion [93], IM3D [48], CAT3D [22], and VideoMV [106]. These works primarily target general objects, but our focus is on human heads. We train multi-view diffusion models on a multi-view head video dataset to obtain more head-specific priors. By using normal maps from FLAME reconstruction as the camera pose condition, we introduce pixel-aligned inductive bias to enable more precise viewpoint control, which is crucial for consistent head Gaussian supervision.

### 3. Preliminaries

#### 3.1. 3D Gaussian Splatting

3D Gaussian Splatting [31] parameterizes a scene using a set of discrete geometric primitives known as 3D Gaussian splats. Each splat is characterized by a covariance matrix  $\Sigma$  centered at location  $\mu$ . The covariance matrices are required to be semi-definite for physical interpretations. It utilizes a parametric ellipsoid definition to construct the covariance matrix  $\Sigma = R S S^T R^T$  using a scaling matrix  $S$  and a rota-

tion matrix  $R$ . These matrices are independently optimized and represented by a scaling vector  $\mathbf{s} \in \mathbb{R}^3$  and a quaternion  $\mathbf{q} \in \mathbb{R}^4$ .  $\mathbf{r} \in \mathbb{R}^{3 \times 3}$  denotes the corresponding rotation matrix of  $\mathbf{q}$ . To represent scene appearance, sphere harmonic coefficients are used to represent the color  $\mathbf{c}$  of each Gaussian. During rendering, a tile-based rasterizer is used to  $\alpha$ -blend all 3D Gaussians overlapping the pixel within a tile. To respect visibility order and avoid per-pixel sorting expenses, splats are sorted based on depth values within each tile before blending.

#### 3.2. Gaussian Avatars

To animate Gaussian splats for head avatars, we use the representation of GaussianAvatars [57] to rig 3D Gaussians using the FLAME mesh. Initially, they bind each triangle of the FLAME identity mesh with a 3D Gaussian and transform the 3D Gaussian based on triangle deformations across different time steps. The splat remains static in the local space of the attached triangle but can dynamically evolve in the global metric space along rotation, translation, and scaling transformations of the binding triangle. For each triangle, they compute the mean position  $\mathbf{T}$  of its three vertices coordinates as the origin of the local space. They define a rotation matrix  $\mathbf{R}$  to depict the orientation of the triangle in the global space, composed of three column vectors derived from the direction vector of one edge, the normal vector of the triangle face, and their cross product. Additionally, they determine triangle scaling  $k$  by calculating the mean length of one edge and its perpendicular in a triangle. The 3D Gaussian is parameterized by the location  $\mu$ , rotation  $\mathbf{r}$ , and anisotropic scaling  $\mathbf{s}$  in the local space of its parent triangle. In the initialization stage, the location  $\mu$  is set as zero, rotation  $\mathbf{r}$  as an identity matrix, and scaling  $\mathbf{s}$  as a unit vector. During rendering, they transform these properties from local to global space by:

$$\mathbf{r}' = \mathbf{R}\mathbf{r} \quad (1)$$

$$\mu' = k\mathbf{R}\mu + \mathbf{T} \quad (2)$$

$$\mathbf{s}' = k\mathbf{s} \quad (3)$$

The adaptive Gaussian densification process operates within local space, where newly added Gaussians inherit binding relations from their original counterparts.

#### 4. Gaussian Avatars Fusion

Given a monocular sequence of RGB images  $\mathcal{I} = \{\mathbf{I}_i\}$  as input, our goal is to reconstruct head Gaussian splats  $\mathcal{O} = \{\mathcal{O}_i\}$  as output. The overall pipeline is shown in Figure 2. To enable animation of the reconstructed avatar with various head poses and expressions, we optimize the Gaussian splats  $\mathcal{O}$  rigged by a parametric head model, *e.g.* FLAME. The challenge is that monocular videos often lack



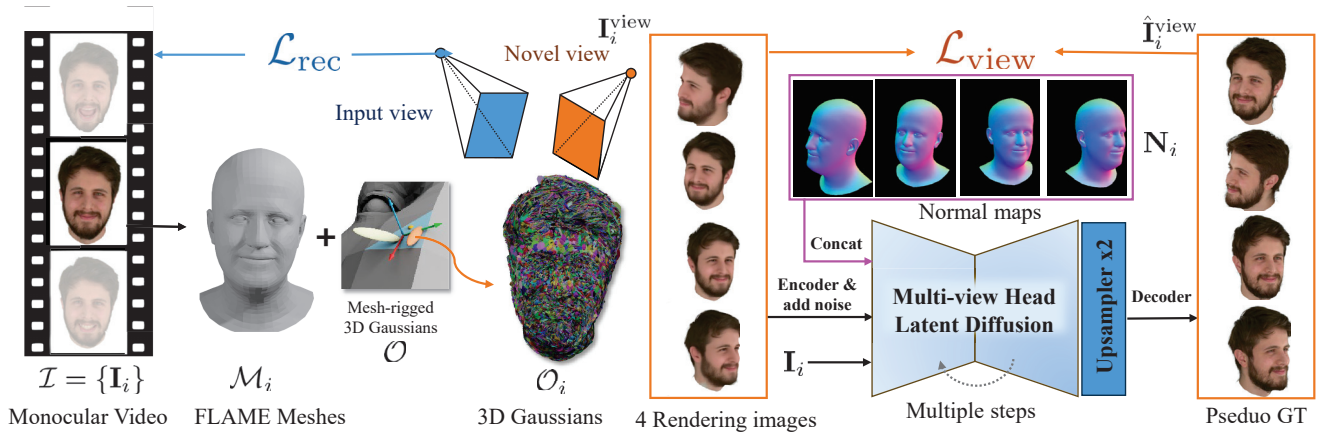


Figure 2. **Pipeline overview.** Given a sequence of RGB images from monocular cameras  $\mathcal{I} = \{\mathbf{I}_i\}$ , our objective is to reconstruct dynamic head avatars by optimizing an animatable Gaussian splatting representation  $\mathcal{O}$ , which is deformed to each frame as  $\mathcal{O}_i$  by the tracked FLAME mesh  $\mathcal{M}_i$  of  $\mathbf{I}_i$ . We optimize  $\mathcal{O}$  by minimizing an input view reconstruction loss  $\mathcal{L}_{\text{rec}}$ , plus a view sampling loss  $\mathcal{L}_{\text{view}}$ .  $\mathcal{L}_{\text{view}}$  compares novel-view renderings of  $\mathcal{O}_i$  from four random viewpoints  $\mathbf{I}_i^{\text{view}}$ , with pseudo ground truths  $\hat{\mathbf{I}}_i^{\text{view}}$ , predicted by a multi-view head latent diffusion model.  $\hat{\mathbf{I}}_i^{\text{view}}$  are generated by iteratively denoising 4-view latents, guided by the input image  $\mathbf{I}_i$  and normal maps  $\mathbf{N}_i$  rendered from  $\mathcal{M}_i$ . A latent upsampler module enhances facial details before decoding the denoised latent into an RGB image.

complete observations of the head. For example, a front-facing video with limited head rotations may provide insufficient information about the face’s sides. This poses difficulties for reconstructing a photorealistic 3D head from partial input, as the optimization over  $\mathcal{O}$  is underconstrained. To address this, we introduce a normal map-guided, multi-view head diffusion model, which is designed to jointly denoise multi-view images conditioned on a single input image and normal maps rendered from FLAME tracking. Once trained, this model serves as a prior to regularize renderings from  $\mathcal{O}$ , filling in unobserved regions and improving the quality of the reconstructed avatar.

#### 4.1. Normal map-conditioned Multi-view Head Latent Diffusion

To address missing regions in monocular videos with limited head coverage, one solution is to distill pre-trained text-to-image diffusion models to regularize novel view renderings. Personalized techniques like Dreambooth [63] allow for identity preservation by customizing diffusion models for specific objects we wish to reconstruct. However, personalized text-to-image diffusion models are not designed to capture the distribution of novel views from a single input image. To overcome this, we propose a novel view diffusion model that generates identity-preserved and appearance-coherent novel views conditioned on an input image. By denoising multiple novel views simultaneously, our approach enhances cross-view consistency. The multi-view head diffusion model is illustrated in Fig. 3.

**Normal Map Conditioning.** To control viewpoint, we leverage normal maps rendered from the FLAME mesh reconstruction at target views as diffusion guidance. The normal map is first encoded into a latent representation via pre-trained VAE [34], matching the dimensionality of the noisy

image latent, and these two latents are then concatenated along the channel dimension. Compared to camera pose embeddings, normal maps offer a more explicit inductive bias for view synthesis by providing pixel-aligned conditioning, which facilitates alignment between the generated images and the conditioning normal maps. Moreover, the FLAME renderings also facilitate the expression accuracy of synthesized novel views.

**Model Architecture.** We train a multi-view diffusion model that takes a single image of a head,  $\mathbf{I}_{\text{cond}}$ , as input and generates multiple output images conditioned on normal maps rendered from desired camera poses using the FLAME reconstruction  $\mathcal{M}$ . Specifically, given  $\mathbf{I}_{\text{cond}}$  and the FLAME mesh  $\mathcal{M}$ , the model learns the joint distribution of  $N$  target images,  $\mathbf{I}_{\text{tgt}}$ , guided by  $N$  normal maps,  $\mathbf{N}_{\text{tgt}}$ , which are rendered from  $\mathcal{M}$  at the target camera poses.

$$p(\mathbf{I}_{\text{tgt}}|\mathbf{I}_{\text{cond}}, \mathbf{N}_{\text{tgt}}) \quad (4)$$

Our model architecture is similar to multi-view diffusion models (MVLDM) [22, 68, 85, 93], which is based on 2D U-Net [62] and attention blocks [83]. As shown in Fig. 3, we use the CLIP image embedding to achieve global control over novel view generation. However, this embedding mainly contains high-level semantic features, lacking the detailed information necessary for accurately capturing the head’s identity and appearance. To address this, we incorporate the VAE latents of  $\mathbf{I}_{\text{cond}}$  directly into the multi-view diffusion model. By applying cross-attention between the multi-view denoised latents and the VAE latent of  $\mathbf{I}_{\text{cond}}$ , we effectively transfer identity-specific details. Additionally, to ensure consistency across views, we aggregate information from the noisy latents across different viewpoints.

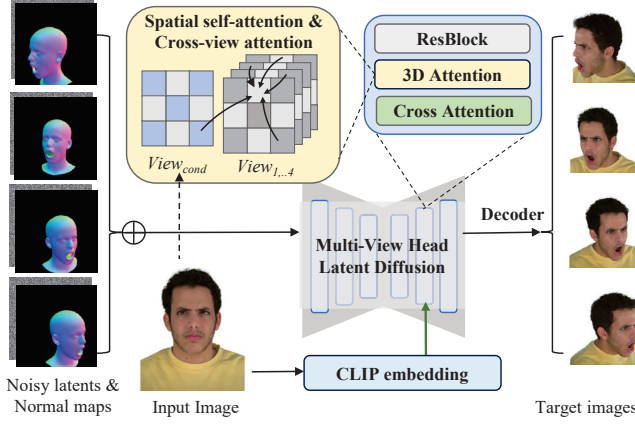


Figure 3. **Multi-view latent head diffusion models.** Given multi-view noisy image latents, we concatenate them with VAE latents of normal maps rendered from FLAME tracking. These combined inputs are processed by a 2D U-Net denoiser with attention blocks. To maintain 3D consistency, 3D attention blocks apply cross-attention across all views, integrating face identity and appearance details from the input image into the denoised latents while exchanging information between noisy latents across views.

## 4.2. Gaussian Avatars Reconstruction with Multi-view Diffusion Priors

We now seek to utilize the multi-view diffusion priors for head Gaussian reconstruction. A commonly used strategy is Score Distillation Sampling (SDS) loss [55], which performs one-step denoising. Due to the stochastic nature of the denoising process introduced by random noise levels and seeds, it contains noisy gradients that disturb 3D optimization. Consequently, it often causes over-saturated appearance issues in synthesized 3D assets. Wu et al. [88] found that after iteratively denoising a noisy image for multiple steps, we can obtain a deterministic output.

**Pseudo-image Ground Truths.** At each iteration, we randomly select the  $i$ -th input frame  $\mathbf{I}_i$  and its FLAME mesh  $\mathcal{M}_i$ . By randomly sampling 4 viewpoints  $\{\phi_j\}_{j=1}^4$ , then we can generate 4 novel views from  $\mathbf{I}_i^{\text{view}} = \mathcal{R}(\mathcal{O}_i, \{\phi_j\})$ , and render normal maps  $\mathbf{N}_i = \mathcal{R}(\mathcal{M}, \{\phi_j\})$  from  $\mathcal{M}_i$ . We can employ normal map-guided multi-view diffusion priors to regularize  $\mathbf{I}_i^{\text{view}}$ . Concretely, we encode  $\mathbf{I}_i^{\text{view}}$  into latent features  $\mathbf{z}$ , which is perturbed with noise  $\epsilon_t$  to obtain noisy features  $\mathbf{z}_t$ . The noise intensity of  $\epsilon_t$  is controlled by the diffusion time step  $t \sim [0.02, 0.98]$ . We then iteratively perform multiple denoising steps of latent diffusion until the final clean latent  $\mathbf{z}_0$  is obtained. To accelerate the generation speed, we adopt the DDIM sampling strategy [69] by running  $t/k$  intermediate steps, which can reduce the denoising steps by  $k = 20$  times. Then  $\mathbf{z}_0$  is decoded back to 4 images  $\{\hat{\mathbf{I}}_i^j\}$  that are served as pseudo supervision.

**Latent Upsampler.** To further enhance the facial details in  $\hat{\mathbf{I}}_i^{\text{view}}$ , we use a pre-trained latent upsampler diffusion model [2] to super-resolve the denoised 4-view latents  $\mathbf{z}_0$ ,

from a resolution of  $32 \times 32$  to  $64 \times 64$ . This super-resolution step allows the pseudo-image ground truths  $\hat{\mathbf{I}}_i^j$  to reach a final resolution of  $512 \times 512$ . We use 10-step DDIM sampling in our latent upsampler inference.

**3D-Aware Denoising.** The rendered views  $\mathbf{I}_i^{\text{view}}$  from a global 3D representation  $\mathbf{O}_i$  introduce 3D-awareness into the denoising process, further enhancing multi-view consistency across pseudo ground truths  $\hat{\mathbf{I}}_i^j$ .

## 4.3. Loss Functions

We supervised the optimization of Gaussian Splats  $\mathcal{O}$  by a combination of loss functions in the following:

$$\mathcal{L} = \mathcal{L}_{\text{img}}(\mathbf{I}^{\text{rec}}, \mathbf{I}) + L_{\text{img}}(\mathbf{I}^{\text{view}}, \hat{\mathbf{I}}^{\text{view}}) + \lambda_{\text{pos}} \mathcal{L}_{\text{pos}} + \lambda_{\text{scale}} \mathcal{L}_{\text{scale}} \quad (5)$$

$L_{\text{img}}$  is defined by a combination of pixel-wise  $L_1$  loss, SSIM loss, and LPIPS loss:

$$L_{\text{img}} = \lambda_1 \mathcal{L}_1 + \lambda_2 \mathcal{L}_{\text{SSIM}} + \lambda_3 \mathcal{L}_{\text{LPIPS}} \quad (6)$$

where  $\lambda_1 = 0.8$ ,  $\lambda_2 = 0.2$ , and  $\lambda_3 = 0.1$ . We also introduce splat position and scale regularization terms to penalize abnormally distributed splats.  $\lambda_{\text{pos}}$  and  $\lambda_{\text{scale}}$  are set to 0.01 and 1 respectively.

## 4.4. Implementation Details

**Multi-view Head Diffusion Model.** It is initialized from Stable Diffusion 2.1 [61] of ImageDream [85] and is trained on eight A100 GPUs over 20,000 iterations, taking approximately 72 hours. The training uses a learning rate of 0.0001 and a batch size of 64. During training, we employ a classifier-free guidance strategy, randomly dropping the input image at a rate of 0.1. The model is trained on the multi-view human head video dataset NeRSemble [36], which contains RGB video sequences from 16 viewpoints, covering both front and side faces. We randomly sample 50,000 timesteps to construct the training dataset.

**Gaussian Avatar Optimization.** The FLAME meshes are initially obtained by VHAP tracker [1] from monocular videos. The animatable Gaussians are optimized with Adam [35] for 6,000 iterations, with learning rates of  $5e-5$ ,  $1.7e-2$ ,  $1e-3$ ,  $2.5e-3$ , and  $5e-2$  for splat position, scaling factor, rotation quaternion, color and opacity respectively. We perform adaptive densification if the position gradients are larger than 0.0002 in every 300 iterations until 5,000 iterations are reached. We also remove Gaussians with opacity less than 0.005. During training, we also fine-tune the FLAME parameters using the learning rates  $1e-6$ ,  $1e-5$ , and  $1e-3$  for translation, joint rotation, and expression coefficients. At each iteration, we randomly sample 4 out of 15 novel viewpoints from the NeRSemble dataset to calculate multi-view diffusion loss.

**Runtime.** Our current implementation of avatar reconstruction takes about 12 hours and uses 32 GB of memory

on a single A6000 GPU. After avatar reconstruction, Our method takes 0.016 seconds to render an image at  $802 \times 550$  resolution, i.e. 62 FPS.

## 5. Experiments

**Datasets.** We conduct head avatar reconstruction experiments on monocular video sequences from the **NeRsemble dataset** [36]. Note that these evaluation sequences were not seen by the multi-view diffusion model. We use monocular videos from the 8-th camera as the input, only capturing the head from the front view. And we use videos from the other 15 views for evaluation. We randomly select 12 sequences from different identities, with durations between 70 and 300 frames, downsampled to a resolution of  $802 \times 550$ . Additionally, we include a **Monocular Video** dataset consisting of 3 monocular videos from the INSTA [105] dataset at  $512 \times 512$  resolution, and 3 sequences of three subjects captured by smartphone at  $1280 \times 720$  resolution.

**Evaluations.** Following previous works [57, 105], we report the average L1 loss, LPIPS, PSNR, and SSIM between renderings and ground truths of the test set. In the NeRsemble dataset, we evaluate head avatar reconstruction and animation quality in two settings: 1) *novel view synthesis*: driving a reconstructed avatar with seen head poses and expressions during training, and rendering it from 15 hold-out viewpoints. 2) *novel expression synthesis*: driving a reconstructed avatar with unseen head poses and expressions during training, and rendering it, rendered from 5 nearby hold-out views, i.e. cameras 6–10. For this dataset, 80% of frames are used for training, and 20% for evaluation. In the Monocular Video dataset, we use around 40% of frames for training and the rest for evaluation. The training sets consist of front-facing frames, while the evaluation set includes obvious head rotations to capture side faces.

**Baselines.** We compare our method against state-of-the-art methods for photo-realistic head reconstruction. *PanoHead* [3] and *SphereHead* [37] use 3D-GAN inversion to reconstruct static heads. They cannot reconstruct animatable avatars. Thus, they cannot be evaluated in novel expression synthesis. *INSTA* [105], *FlashAvatar* [89], and *GA* [57] optimize photo-realistic head avatars represented by Instant NGP or Gaussian Splatting. They overfit avatars to monocular observations. *P4D-v2* [18] and *GAGAvatar* [14] reconstruct animatable 3D avatars from single images, that can then be reenacted with novel expressions.

### 5.1. Head Avatar Reconstruction

**NeRsemble.** As shown in Fig. 4, front-facing monocular videos lack sufficient side-face information. *INSTA*, *FlashAvatar*, and *GA* can only reconstruct observed regions and leave unobserved areas unconstrained. This often leads to artifacts in extreme hold-out views. *PanoHead* [3] and

| Method           | Novel Views |        |        | Novel Expressions |        |        |
|------------------|-------------|--------|--------|-------------------|--------|--------|
|                  | LPIPS ↓     | PSNR ↑ | SSIM ↑ | LPIPS ↓           | PSNR ↑ | SSIM ↑ |
| PanoHead [3]     | 0.171       | 17.40  | 86.11  | N/A               | N/A    | N/A    |
| SphereHead [37]  | 0.174       | 16.94  | 85.98  | N/A               | N/A    | N/A    |
| INSTA [105]      | 0.262       | 15.87  | 77.02  | 0.165             | 19.46  | 84.91  |
| FlashAvatar [89] | 0.247       | 16.94  | 81.05  | 0.145             | 21.37  | 86.08  |
| GA [57]          | 0.218       | 17.51  | 81.66  | 0.138             | 21.63  | 87.00  |
| P4D-v2 [18]      | 0.161       | 16.82  | 85.44  | 0.142             | 18.72  | 86.63  |
| GAGAvatar [14]   | 0.183       | 21.20  | 80.72  | 0.177             | 21.32  | 80.44  |
| Ours             | 0.125       | 20.88  | 88.91  | 0.087             | 24.12  | 90.66  |

Table 1. Quantitative comparisons on dynamic avatar reconstruction and animation from monocular videos. Results are obtained by the average of twelve sequences of different subjects on the **NeRsemble dataset**. **Best** and **2nd-best** are highlighted.

|                  | L1 ↓   | LPIPS ↓ | PSNR ↑ | SSIM ↑ |
|------------------|--------|---------|--------|--------|
| INSTA [105]      | 0.0407 | 0.177   | 21.11  | 85.04  |
| FlashAvatar [89] | 0.0376 | 0.145   | 21.13  | 85.93  |
| GA [57]          | 0.0263 | 0.109   | 22.55  | 88.61  |
| P4D-v2 [17]      | 0.0317 | 0.131   | 20.95  | 86.82  |
| GAGAvatar [14]   | 0.0282 | 0.108   | 22.34  | 85.38  |
| Ours             | 0.0229 | 0.090   | 23.16  | 89.76  |

Table 2. Quantitative comparisons on dynamic avatar reconstruction on **Monocular Video dataset**. Results are obtained by **hold-out expression frames** from six sequences of INSTA and smartphone capture. Note that **novel view renderings cannot be evaluated** due to the absence of ground truths in single-view captures.

*SphereHead* [37] fail to generate plausible renderings in extreme novel views. *P4D-v2* [18] exhibits blurriness and artifacts in the mouth and ears, while *GAGAvatar* [14] suffers from identity drift due to weak constraints in less observed regions. In contrast, our method leverages multi-view diffusion priors to improve novel view synthesis, effectively completing missing regions and enhancing photorealism while preserving identity and appearance consistency. As shown in Tab. 1, our approach outperforms all baselines in LPIPS and SSIM and ranks second in PSNR on the task of novel view synthesis. Interestingly, novel view constraints also benefit novel expression synthesis, resulting in superior performance across all metrics.

**Monocular Videos.** We also provide the comparisons on the Monocular Video dataset. Since single-view datasets lack ground-truth novel views, we evaluate animation results by applying expression parameters from hold-out frames. As shown in Tab. 2, our method consistently surpasses others in all metrics. The qualitative results are presented in Fig. 5. While all methods can well reconstruct the input view (top right), *GA* [57] struggles to generate plausible novel views (bottom right), particularly for unseen face regions. It is also prone to artifacts and inconsistencies in novel pose animations. *P4D-v2* [18] and *GAGAvatar* [14]



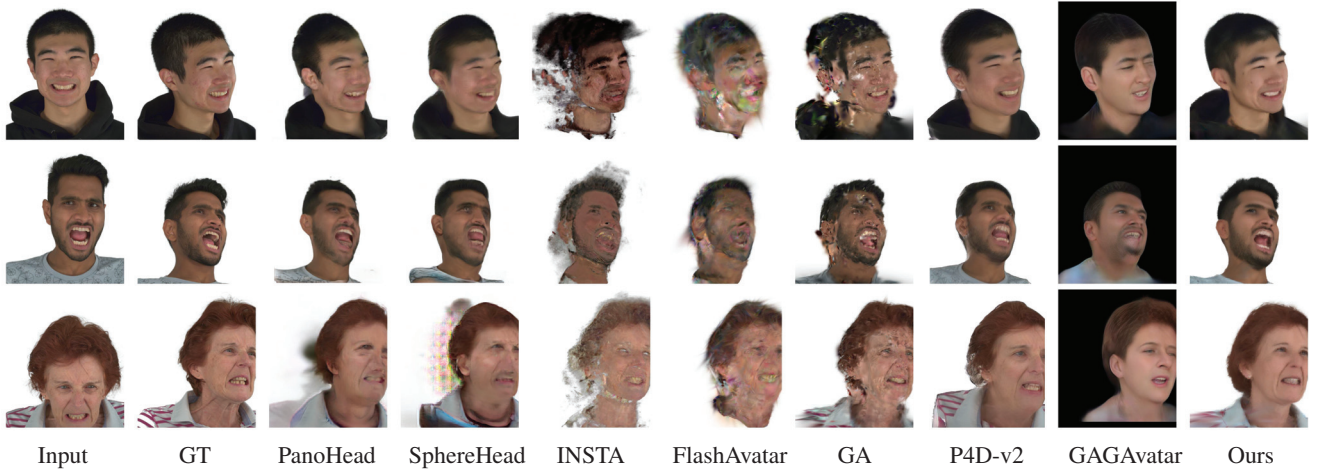


Figure 4. **Novel view synthesis from monocular videos on the NeRSemble dataset.** Compared to state-of-the-art methods, our approach reconstructs unseen side facial regions in the inputs, better preserves facial identities, and consistently produces more favorable renderings from hold-out views.

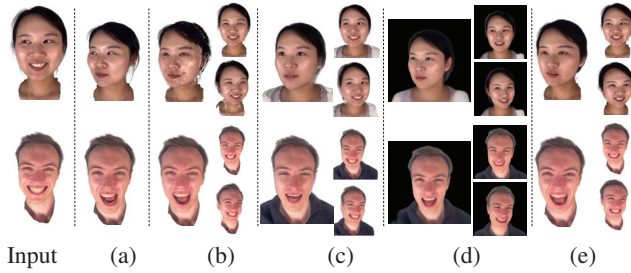


Figure 5. **Head avatar reconstruction from monocular videos captured on commodity devices.** (a) Ground truth of novel expressions; (b) GA; (c) P4D-v2; (d) GAGAvatar; (e) Ours. For each method, we display the fitting results of the input frame (top right) and novel view renderings of the input frame (bottom right). Given a front-facing sequence with limited head poses, all methods can accurately reconstruct the observed frames. However, without effective priors to constrain unobserved regions, GA struggles to generalize to novel views and poses. P4D-v2 and GAGAvatar exhibit limitations in capturing complicated facial expressions or fine-grained details such as wrinkles.

have limited capacity to represent exaggerated expressions or fine details such as wrinkles.

## 5.2. Ablation Studies

We conduct detailed ablation studies to evaluate diffusion prior choices for novel view constraints, confirming the effectiveness of each design in our face-specific multi-view diffusion prior learning for 3D head reconstruction from monocular videos. We use six sequences from the NeRSemble dataset. The results are presented in Fig. 6, and Tab. 3. Please refer to the supplementary material for more implementation details of ablation studies.

### What is the effect of multi-view head diffusion priors?

An alternative is to use pre-trained text-to-image diffusion models Stable Diffusion [61]. Another alternative is to fine-

| Method              | Novel Views  |              |              | Novel Expressions |              |              |
|---------------------|--------------|--------------|--------------|-------------------|--------------|--------------|
|                     | LPIPS ↓      | PSNR ↑       | SSIM ↑       | LPIPS ↓           | PSNR ↑       | SSIM ↑       |
| No diffusion        | 0.207        | 18.47        | 82.69        | 0.129             | 22.74        | 88.13        |
| pretrained SD       | 0.196        | 16.99        | 83.47        | 0.149             | 21.37        | 87.81        |
| personalized SD     | 0.196        | 16.89        | 85.98        | 0.147             | 21.51        | 89.52        |
| Ours, w/o pseudo GT | 0.178        | 19.57        | 84.73        | 0.136             | 22.41        | 87.59        |
| Ours, pose cond.    | 0.131        | 19.69        | 87.63        | 0.087             | 23.68        | 90.37        |
| Ours, raymap cond.  | 0.150        | 18.56        | 86.70        | 0.095             | 22.84        | 89.73        |
| Ours, w/o latent×2  | 0.134        | 21.48        | 89.53        | 0.087             | 25.07        | 91.75        |
| <b>Ours final</b>   | <b>0.118</b> | <b>21.82</b> | <b>89.87</b> | <b>0.079</b>      | <b>25.39</b> | <b>92.02</b> |

Table 3. **Ablation Studies on different types of diffusion priors.** Results are obtained from the average of six sequences of different subjects from the NeRSemble [36] dataset.

tune Stable Diffusion on RGB frames from the input video, obtaining personalized image diffusion priors. We implement both variants using normal map guidance of ControlNet [98]. In Fig. 6 (b), pre-trained text-to-image priors often produce renderings that deviate from the original identity. In Fig. 6 (c), personalized diffusion priors improve identity preservation but struggle with appearance consistency, as they lack input image information to hallucinate novel views. Our approach learns to jointly generate multiple novel views conditioned on the input image, thus achieving higher realism and view consistency in both identity and appearance.

### What is the effect of normal map conditioning for multi-view diffusion models?

Our multi-view head diffusion models condition on normal maps of FLAME reconstruction, which are pixel-aligned with the target novel-view images. Fig. 6 (d) and (e) reflect that using camera pose embedding and ray map conditioning could introduce obvious misalignment errors in synthesizing pseudo-image ground truths, leading to blurred 3D Gaussian renderings.

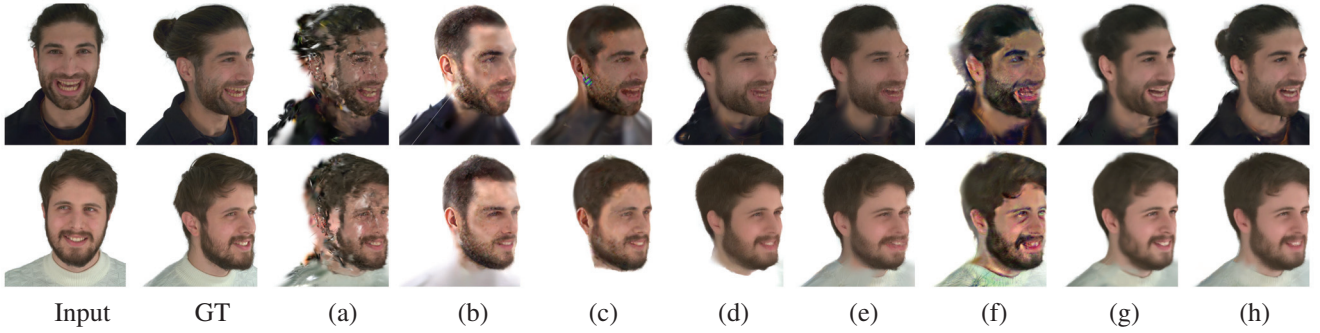


Figure 6. **Ablation studies on different types of diffusion priors.** Comparisons between method variants of (a) No diffusion; using (b) Pretrained Stable Diffusion; (c) Personalized Stable Diffusion; (d) Pose-conditioned multi-view diffusion; (e) Raymap-conditioned multi-view diffusion; (f) Our multi-view diffusion using Score Distillation Sampling (SDS) loss; (g) Ours without latent upsampler  $\times 2$ ; (h) Our final model. Our normal map-conditioned multi-view diffusion priors enable more photo-realistic novel views with identity and appearance consistency, by constraining novel views using pseudo-image ground truths, which are decoded from iteratively denoised latents followed by the latent upsampler.

**What is the effect of iteratively denoised images as pseudo-ground-truths?** We can instead use Score Distillation Sampling (SDS) loss to constrain multi-view renderings. As shown in Fig. 6 (f), vanilla one-step SDS loss has appearance oversaturation issues in the face region.

**What is the effect of latent upsampler module?** Through Fig. 6 (g) vs. (h), we can see that the upsampler module significantly sharpens and enhances facial appearance details.

### 5.3. Multi-view Head Diffusion Models

We compare normal map conditioning to alternatives in our multi-view head diffusion models, including pose embedding and Plücker ray maps [54]. Results in Tab. 4 and Fig. 7 show that FLAME normal maps can achieve more superior facial alignment, with  $\times 2.56$  lower  $l_1$  error and 7.73% higher SSIM.

| Method         | $l_1 \downarrow$ | LPIPS $\downarrow$ | PSNR $\uparrow$ | SSIM $\uparrow$ |
|----------------|------------------|--------------------|-----------------|-----------------|
| pose embedding | 0.123            | 0.184              | 16.52           | 76.12           |
| ray map        | 0.137            | 0.198              | 16.05           | 75.71           |
| normal map     | <b>0.048</b>     | <b>0.119</b>       | <b>19.67</b>    | <b>83.85</b>    |

Table 4. **Ablation studies on different camera conditioning in multi-view head diffusion.**

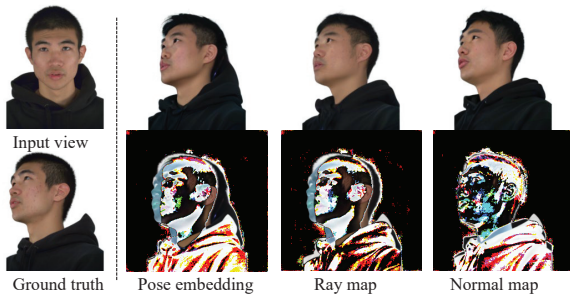


Figure 7. **Ablation studies on different camera conditioning in multi-view head diffusion.** We show predictions in the first row and  $l_1$  difference maps between the ground truth and predictions.

### 5.4. Limitations and Future Work

While our work has shown promising results in dynamic avatar reconstruction from monocular videos captured on studio setups or commodity devices, there are limitations in our current method. First, we do not explicitly separate the material and appearance of heads, which could enable re-lighting applications [33, 65]. Second, optimizing head Gaussians using iteratively updated pseudo ground-truths from diffusion models is time-consuming. We plan to explore real-time 4D avatar reconstruction with feed-forward large reconstruction models [30, 38, 97]. Lastly, the quality of our avatar reconstruction and animation is limited by the expressiveness of current parametric head models, which lack detailed hair geometry and animation. Future work could extend Gaussian head avatars to incorporate fine-grained hair modeling and animation. [45, 95].

## 6. Conclusion

In this work, we present a novel method to reconstruct photo-realistic head avatars from monocular videos to push the frontier of avatar fidelity from commodity devices. Due to limited observation and coverage of human heads, Gaussian reconstruction from monocular videos is inherently under-constrained. To address this challenge, we introduce multi-view diffusion priors that jointly constrain photorealism across multiple views rendered from Gaussian splats. We obtain these priors by designing a multi-view head diffusion model, fine-tuned on a multi-view head video dataset to generate novel views from a single image, conditioned on rendered normal maps from FLAME tracking. To mitigate over-saturated appearances, we use iteratively denoised images as pseudo-ground truths. A pre-trained latent upsampler diffusion model further enhances facial details. By reducing data capture requirements for avatar creation, our approach has the potential to unlock new opportunities in immersive VR/AR applications and products.



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